Fiscal Year:	FY 2015	**Task Last Updated:**	FY 07/25/2014
PI Name:	Kelton, Kenneth F. Ph.D.		
Project Title:	NASA Research under ESA-Based Investigations THERMOLAB and ICOPROSOL		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	kfk@wuphys.wustl.edu	**Fax:**	FY 314 935-6219
PI Organization Type:	UNIVERSITY	**Phone:**	314-935-6228
Organization Name:	Washington University		
PI Address 1:	Department of Physics, Campus Box 1105		
PI Address 2:	One Brookings Drive		
PI Web Page:			
City:	St. Louis	**State:**	MO
Zip Code:	63130	**Congressional District:**	1
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
Start Date:	10/01/2010	**End Date:**	12/31/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
Contact Monitor:	Downey, James1 Patton	**Contact Phone:**	
Contact Email:	patton.downey@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Extended to 12/31/2015 per NSSC information (Ed., 7/8/15) NOTE: To be extended to 9/30/2015 per PI (Ed., 7/10/13)		
Key Personnel Changes/Previous PI:	Kenneth Kelton is U.S. Team Member on ESA-sponsored projects, “Thermophysical properties and solidification behaviour of undercooled Ti-Zr-Ni liquids showing an icosahedral short-range order (ICOPROSOL)," and "Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS)."		
COI Name (Institution):			
Grant/Contract No.:	NNX10AU19G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

Funding is for Dr. Kelton's role as Team Member on the subprojects described below, associated with ESA-sponsored main projects as noted.
Main project: Thermophysical and structural properties and solidification behavior of supercooled glass forming liquids as a member of THERMOLAB; the Science Coordinator is Prof. Dr. Hans Fecht, Ulm University, Institute of Micro and Nanomaterials, Albert-Einstein-Allee 47, Ulm Germany. Similar measurements on undercooled Ti-Zr-Ni liquids showing an icosahedral short-range order (ICOPROSOL); the Science Coordinator is Dr. Dirk Holland-Moritz, Deutsches Zentrum für Luft- und Raumfahrt (DLR), Institut für Materialphysik im Weltraum, Linder Höhe, Köln, Germany.

Subprojects:

1. Ground-Based Experimental Program: Synchrotron X-ray studies of liquid structures are made at the Advanced Photon Source using the ESL chamber at Washington University (WU-BESL), which has been optimized for scattering studies. These are correlated with thermophysical property data obtained on ground with WU-BESL and using the MSL-EML on the ISS. Terrestrial maximum undercooling and induction time to recalescence data are obtained in ESL, where stirring ensures that nucleation is not diffusion controlled. This data will allow an evaluation of diffusion in the upcoming MSL-EML ISS undercooling studies.

2. Thermophysical Property Measurements: ISS. Specific heat, density, surface tension and viscosity measurements will be made of supercooled Ti-Zr-Ni liquids with different Ni concentrations as a function of the temperature using the MSL-EML facility. These will be correlated with x-ray synchrotron diffraction studies of the supercooled liquids and ground-based measurements made using WU-BESL The purpose is to determine the existence and nature of liquid/liquid phase transitions that may be associated with chemical and topological ordering and a possible connection with liquid fragility.

3. Effect of diffusion on nucleation: The importance of long-range diffusion on nucleation will be evaluated using the MSL-EML facility located on the ISS. The critical items of interest are the maximum undercooling and the induction time to recalescence in isothermal holds at temperatures below the liquidus temperature. Together with the results from the thermophysical property measurements and an MHD convection model results this will allow us to analyze how the maximum undercooling is influenced by fluid flow. These results of the micro-g experiments will be compared with the results of ground-based ESL measurements and with predictions from Kelton’s coupled-flux model for nucleation.

Main project: Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS); Science Coordinator is Professor Hans-Joerg Fecht, Ulm University, Ulm, Germany

Subprojects:

1. Ground Based Experiment Program. Measurements of the density, surface tension and viscosity and synchrotron x-ray diffraction measurements of the liquid structure in the supercooled state using WU-BESL, Ground-based measurements of the maximum undercooling and induction time to recalescence in isothermal holds will also be made using WU-BESL. These will be used to evaluate the nature of ordering on phase transitions in the supercooled liquid and fragility.

2 Long Duration Microgravity Experiments. Measurements of thermophysical properties, specific heat (modulated measurements), surface tension, viscosity and density, and also electrical resistivity will be made as a function of temperature using the MSL-EML on the ISS. These will be coupled with x-ray diffraction measurements of the liquid structure factor (WP 2) and ground based studies of the density and viscosity to study the role of ordering on novel liquid/liquid phase transitions and to determine the nature of these transitions. Maximum undercooling and time to recalescece measurements as a function of temperature and stirring will be compared with terrestrial ESL measurements to assess the importance of diffusion on nucleation, allowing Kelton's coupled flux model for nucleation to be evaluated experimentally.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Product advancement in industries such as information technology, aerospace, automotive, and basic and new materials manufacturing requires advanced materials development. In the past, this was often carried out through a laborious process of guided trial and error. The vast increases in computing power have now opened the door for realistic modeling approaches, which has the potential to greatly improve the efficiency of materials development. In the US, this has become a central focus of research in materials science, encapsulated in the Materials Genome Initiative. However, progress is frequently hampered by the lack of relevant thermophysical property data needed for the modeling. Further, key processes such as crystal nucleation from the liquid or other phase transitions within the liquid, are often poorly understood, making it difficult to realistically model them. In many cases, the experimental data and a deeper understanding of some processes cannot be obtained from terrestrial experiments due to interactions with sample containers at the temperatures of interest. Even containerless based methods, such as electrostatic levitation, cannot be used to accurately investigate these. The ISS experiments planned under the support of this grant will allow these data to be obtained.

The focus of NNX10AU19G is to support ground-based structural, density, and viscosity studies of supercooled liquids, and to support an ISS-flight-based study of specific heat, emissivity, thermal conductivity, viscosity, and nucleation processes in a micro-gravity environment for QUASI. As a member of the THERMOLAB team, QUASI will carry out measurements on Vit106a, a Zr-based liquid that easily forms a metallic glass. QUASI is the principle investigation for Vit106, a closely related Zr-based BMG forming alloy to that of Vit106a. Samples for Batch 2 measurements on the ISS are currently under consideration; a QUASI sample, Zr47Cu46Al6, is one of the potential samples. Samples under consideration for Batch 3 that are QUASI samples are ZrPt, ZrCuNiAlTi (in collaboration with THERMOLAB), and TiZrNi (in collaboration with ICOPROSOL). Samples of Zr48Cu46Al6 and Zr80Pt20 are being prepared for sample parameter definition studies in September 2014 and parabolic flight planned for fall 2014.
This past year, this grant has provided partial support for the PI, a Research Scientist, and two graduate students. A brief summary of key ground-based activities during the past year is given here. These investigations are aimed at obtaining supporting information that will be used to help design the ISS experiments and to help interpret the results.

Task Progress:

Activities directly linked with upcoming ISS experiment-- Many activities were carried out over the past year that are directly focused on supporting the ISS experiments. These include:

Support of an Internal Working Group meeting, December 12 and 13, 2013. These meetings are required for ISS experimental planning and exchange of project information within the THERMOLAB and ICOPROSOL teams, as well as with other teams. Because of the uncertainty of available funds due to the US Government shutdown, I joined this meeting by Skype.

Preparation of samples needed for parameter determination by the Microgravity User Support Center (MUSC) – The ISS-EML and the parabolic flight experiments require that the coupling parameters for each sample be known. We have delivered Ti-Zr-Ni samples and are in the process of delivering ZrPt samples for these measurements. The sample preparation methods used are identical to those that will be used to prepare the flight samples. Samples will be delivered in the early fall for parabolic flight studies. Measurement of the evaporation rates of proposed samples for flight investigation – Evaporation rates are required to calculate coil coating and dust production during processing in the MSL-EML on the ISS. Previously, using the Washington University Beamline ESL (WU-BESL), we measured the evaporation rates for the flight samples Vit 106 and Vit 106a. Over the past year, we have made measurements on samples that have been proposed for Batch 2: CU46Zr54, Cu50Zr50, Cu47Zr47Al6, Zr64Ni36, Ni59.5Nb40.5, Zr52.5Ti5Cu17.9Ni14.6Al10, and Zr50.75Cu36.22Ni4.03Al9.

Ground-based activities to obtain supporting information for ISS experiments-- Some of the ground-based activities were directed at providing additional data that will help design the ISS experimental studies and help interpret the results. Key results from these studies are listed below:

Correlation between Liquid Specific Volume and Glass Formability – Density measurements were made with WU-BESL as a function of temperature for seven supercooled liquids that easily form metallic glasses. These show that on average the difference in the magnitudes of thermal expansion coefficients in the glass and liquid is less for the better glass formers. The thermal expansion coefficients for liquids arise from the anharmonicity of the interatomic potential, which is similar to that in the glass, and from structural ordering of the liquid. These results suggest that the better glass-forming liquids have less structural ordering with decreasing temperature. This is consistent with a large degree of structural order already being present at high temperature in the best glass-forming liquids.

Specific heat and total hemispherical emissivity of bulk metallic glasses and liquids – Few data exist for the specific heat, Cp, of supercooled liquids; this is one focus of the planned ISS experiments, where it is possible to measure Cp accurately using modulated calorimetry. To refine the protocol for the ISS measurements and to have complementary data to those from the flight experiments, ground-based specific measurements have been made on Vit 106, one of the samples in the first batch to be processed on the ISS. By combining ESL data from the glass with more direct measurements of Cp made in a differential scanning calorimeter, the hemispherical emissivity can be determined, allowing Cp to be measured across a broad temperature range that encompasses the equilibrium and supercooled liquid, as well as the glass. To our knowledge, with the exception of Zr46.75Ti8.25Cu7.5Ni10Be2.5, this is the first complete set of specific heat data for a metallic glass and its corresponding liquid.

Invalidity of the Ehrenfest Relation for High Temperature Liquids – For liquids and glasses Ehrenfest argued that the location of the first peak in the structure factor, S(q1), is proportional to the negative cube root of the specific volume, V. Recently, a paper in Nature Materials reported that for glasses the exponent is -0.433 instead of -0.333. Based on a comprehensive survey of dozens of liquid metals, we have recently discovered the same result there. The authors of the study for the glasses argued that this occurs because the glass is a network with fractal dimension of Df = 1/0.433 = 2.31, which is similar to the measured fractal dimension for quasicrystals. That the same value would be found at high temperatures suggests that the liquids would be similarly ordered. Since it is known that glasses are better ordered than liquids and that structural ordering occurs in the liquid with decreasing temperature, this appears unlikely. The meaning of the exponent, then, is a subject for further theoretical study.

Anomalous Thermal Expansion of the First Peak in the Pair Correlation Function – Based on synchrotron X-ray scattering measurements we have demonstrated that in many cases (but not all) the distances between nearest neighbor atoms increase with decreasing temperature, giving what amounts to a negative thermal expansion coefficient. We show that this is primarily due to an increase in the number of neighboring atoms with decreasing temperature, which weakens the bond strength and allows the atoms to move apart. This occurs in all liquids studied and is consistent with molecular dynamics simulations. The weighting factors for the different chemical pair correlations, however, determine whether the observed local expansion is positive or negative.

Liquid Structural Fragility – Liquids are often described in terms of their kinetic fragility, based on the temperature dependence of the shear viscosity. For strong liquids, such as SiO2, the activation energy for flow is weakly dependent on temperature. However, for fragile liquids such as o-terphenyl, the activation energy is small at high temperatures, but decreases rapidly upon approaching the glass transition temperature, Tg. Based on synchrotron X-ray scattering studies of metallic glasses and the corresponding liquids, we have identified a structural origin of fragility, based on the rate of structural ordering near the glass transition temperature. A link between structure and dynamics has long been argued from theoretical and modeling considerations. While there have also been hints of this link in experimental data, our studies are the most direct confirmation to date. This work was recently accepted for publication in Nature Communications.

Universal Viscosity Curve – Typically the logarithm of the shear viscosity of liquids is plotted as a function of the product of inverse temperature and the glass transition temperature. However, we have discovered a new scaling temperature, Tcoop, that is more fundamental than Tg. It corresponds to the onset of cooperative atom motion within the liquid. For the metallic liquids studied Tcoop is approximately 10 percent above the melting temperature. Further, we have demonstrated that by scaling the inverse temperature to Tcoop instead of Tg and the viscosity to the extrapolated infinite temperature value, all viscosity data that we have measured fall onto a universal curve. We have found that Tcoop is approximately equal to 2Tg, indicating that the glass transition is a final realization of the cooperative behavior that starts at a much higher temperature in the equilibrium liquid.

Papers submitted or in press during the reporting period:

M. Blodgett, T. Egami, Z. Nussinov, K. F. Kelton, Unexpected Universality in the Viscosity of Metallic Liquids, submitted to Science July, 2014.

J. C. Bendert, C. E. Pueblo, S. Veligati, N. A. Mauro and K. F. Kelton, “Temperature Calibration of Optical Pyrometry

	in Containerless Systems using Differential Scanning Calorimetry–Applied to Cu100-xZrx(x=45-50),” International Journal of Thermophysics (2014, in press). R. C. Bradshaw, R. W. Hyers, J. R. Rogers, T. J. Rathz, G. W. Lee, A. K. Gangopadhyay and K. F. Kelton, “Non-contact Techniques for Measurements of Density: Data for a Quasicrystal-Forming Ti-Zr-Ni Alloy,” International Journal of Thermophysics (2014, in press). M. Blodgett, A. K. Gangopadhyay and K. F. Kelton, “Thermal Evaporation Measurements for Quasicrystal (TiZrNi) and Glass Forming (Vit 106 and 106a) Liquids,” International Journal of Thermophysics (2014, in press).
Bibliography Type:	Description: (Last Updated: 05/20/2025)
Articles in Peer-reviewed Journals	Mauro NA, Blodgett M, Johnson ML, Vogt AJ, Kelton KF. "A structural signature of liquid fragility." Nature Communications. 2014 Aug 6;5:4616. http://dx.doi.org/10.1038/ncomms5616 ; PubMed PMID: 25098937 , Aug-2014
Articles in Peer-reviewed Journals	Gangopadhyay AK, Blodgett ME, Johnson ML, Vogt AJ, Mauro NA, Kelton KF. "Thermal expansion measurements by X-ray scattering and breakdown of Ehrenfest's relation in alloy liquids." Applied Physics Letters. 2014 May;104(19):191907. http://dx.doi.org/10.1063/1.4876125 , May-2014
Articles in Peer-reviewed Journals	Gangopadhyay AK, Blodgett ME, Johnson ML, McKnight J, Wessels V, Vogt AJ, Mauro NA, Bendert JC, Soklaski R, Yang L, Kelton KF. "Anomalous thermal contraction of the first coordination shell in metallic alloy liquids." Journal of Chemical Physics. 2014 Jan;140(4):044505. http://dx.doi.org/10.1063/1.4861666 , Jan-2014
Articles in Peer-reviewed Journals	Mauro NA, Vogt AJ, Johnson ML, Bendert JC, Soklaski R, Yang L, Kelton KF. "Anomalous structural evolution and liquid fragility signatures in Cu-Zr and Cu-Hf liquids and glasses." Acta Materialia. 2013 Nov;61(19):7411-21. http://dx.doi.org/10.1016/j.actamat.2013.08.047 , Nov-2013
Articles in Peer-reviewed Journals	Bendert JC, Kelton KF. "Containerless measurements of density and viscosity for a Cu48 Zr52 liquid." International Journal of Thermophysics. 2014 Oct;35(9-10):1677-86. http://dx.doi.org/10.1007/s10765-014-1664-7 , Oct-2014
Articles in Peer-reviewed Journals	Bendert JC, Pueblo CE, Veligati S, Mauro NA, Kelton KF. "Temperature calibration of optical pyrometry in containerless systems using differential scanning calorimetry: Cu100-xZrx (x=45–50)." International Journal of Thermophysics. 2014 Oct;35(9-10):1687-96. http://dx.doi.org/10.1007/s10765-014-1660-y , Oct-2014
Articles in Peer-reviewed Journals	Bradshaw RC, Hyers RW, Rogers JR, Rathz TJ, Lee GW, Gangopadhyay AK, Kelton KF. "Non-contact techniques for measurements of density: Data for a quasicrystal-forming Ti-Zr-Ni alloy." International Journal of Thermophysics. In press as of September 2014. , Sep-2014
Articles in Peer-reviewed Journals	Blodgett M, Gangopadhyay AK, Kelton KF. "Thermal evaporation measurements for quasicrystal (TiZrNi) and glass forming (Vit 106 and 106a) liquids." International Journal of Thermophysics. In press as of September 2014. , Sep-2014